Electrochemistry

1. Which of the following solutions of KCI has the lowest value of equivalent conductance?
1) 1 M
2) 0.1 M
3) 0.01 M
4) 0.001 M

2. The specific conductance in $ohm^{-1}\ cm^{-1}$ of four electrolytes P, Q, R, and S are given in brackets:

$P\left(5.0\times10^{-5}\right)$ $Q\left(7.0\times10^{-8}\right)$

$R\left(1.0\times10^{-10}\right)$ $S\left(9.2\times10^{-3}\right)$

The one that offers the highest resistance to the passage of electric current is:
1) P
2) Q
3) R
4) S

3. At 25°C specific conductivity of a normal solution of KCl is 0.002765 $\Omega^{-1}\,cm^{-1}$ The resistance of cell is 400 ohms. The cell constant is
1) 0.815
2) 1.016
3) 1.106
4) 2.016

4. If the conductance and specific conductance of a solution is one then its cell constant would be
1) 1
2) Zero
3) 0.5
4) 4

5. The molar conductance of NaCl, HCI and CH_3COONa at infinite dilution are 126.45, 426.16 and 91 $ohm^{-1}\ cm^2\ mol^{-1}$ respectively. The molar conductance of CH_3COOH at infinite dilution is:
1) 201.28 $ohm^{-1}\ cm^2\ mol^{-1}$
2) 390.71 $ohm^{-1}\ cm^2\ mol^{-1}$
3) 698.28 $ohm^{-1}\ cm^2\ mol^{-1}$
4) 540.48 $ohm^{-1}\ cm^2\ mol^{-1}$

6. The resistance of 0.1 N solution of an electrolyte was found to be 210 ohm at 298 K using a conductivity cell of cell constant 0.66 cm^{-1}. The equivalent conductance of solution is:
1) 314.28 mho $cm^2\ eq^{-1}$
2) 3.14 mho $cm^2\ eq^{-1}$
3) 314.28 mho $cm^2\ eq^{-1}$
4) 3.14 mho $cm^2\ eq^{-1}$

7. The specific conductance of a salt of 0.01 M concentration is $1.061\times10^{-4}\ S\ cm^{-1}$. Molar conductance of the same solution will be

1) $1.061\times10^{-4}\,S\,cm^2\,mol^{-1}$
2) $1.061\,S\,cm^2\,mol^{-1}$
3) $10.61\,S\,cm^2\,mol^{-1}$
4) $106.1\,S\,cm^2\,mol^{-1}$

8. The increase in equivalent conductance of a weak electrolyte with dilution is due to:

1) Increase in degree of dissociation and decrease in ionic mobility
2) Decrease in degree of dissociation and decrease in ionic mobility
3) Increase in degree of dissociation and increase in ionic mobility
4) Decrease in degree of dissociation and increase in ionic mobility

9. At infinite dilution, the equivalent conductance of CH_3COONa, HCl and CH_3COOH are 91, 426 and 391 mho $cm^2\ eq^{-1}$ respectively at 25°C. The eq. conductance of NaCl at infinite dilution will be:

1) 126
2) 209
3) 391
4) 908

10. The conductivity of a saturated solution of $BaSO_4$ is $3.06\times10^{-6}\ ohm^{-1}\ cm^{-1}$ and its molar conductance is 1.53 $ohm^{-1}\ cm^2\ mol^{-1}$. The K_{sp} of $BaSO_4$ will be

1) 4×10^{-12}
2) 2.5×10^{-9}
3) 2.5×10^{-13}
4) 4×10^{-6}

11. The emf of the cell $Tl(s)|Tl^+(0.0001\ M)||Cu^{2+}(0.01\,M|Cu(s)$ is 0.83 V The emf of this cell will be increased by:

1) Increasing the concentration of Cu^{+2} ions
2) Decreasing the concentration of Tl^+
3) Increasing the concentration of both
4) (1) and (2) both

12. The electrode potential of a hydrogen electrode dipped in a solution of pH = 1 is:

1) 0.059 V
2) 0.00 V
3) – 0.059 V
4) 0.59 V

13. E° for $F_2+2e^- \longrightarrow 2F^-$ is 2.8 V, E° for $1/2F_2+e^-$ is?

1) 2.8 V
2) 1.4 V
3) – 2.8 V
4) – 1.4 V

14. How much will the potential of a hydrogen electrode change when its solution initially at pH = 0 is neutralised to pH = 7?

1) increases by 0.059 V
2) decreases by 0.059 V
3) increases by 0.41 V
4) decreases by 0.41 V

15. Consider the reaction, $Cl_2(g)+2Br^-(aq)\rightarrow 2Cl^-(aq)+Br_2(\ell)$. The emf of the cell when $\left[Cl^-\right]=\left[Br^-\right]=0.01$ M and Cl_2 gas at 0.25 atm pressure will be (E0 for the above reaction is 0.29 V)

1) 0.54 V
2) 0.272 V
3) 0.29 V
4) – 0.29 V

16. The Nernst equation E = E° -RT/nF InQ indicates that the Q will be equal to the equilibrium constant Kc when:

1) $E=E^0$
2) $RT/nF=1$
3) $E=Zero$
4) $E^0=1$

17. Calculate the cell potential (in V) if $\Delta G=-96.5\,kJ/mol$ and $n=1$:

1) 0.1
2) 1
3) 10
4) 100

18. The E° at 25°C for the following reaction is 0.55 V. Calculate the AG° in kJ:

$$4BiO^+(aq)+3N_2H_5^+\rightarrow 4Bi(s)+3N_2(g)+4H_2O(l)+7H^+(aq)$$

1) – 637
2) – 424
3) – 106
4) – 318.5

19. Use the following E° for the electrode potentials, and calculate $\Delta G°$ in kJ for the indicated reaction:

$$5Ce^{4+}(aq)+Mn^{2+}(aq)+4H_2O(l)\rightarrow 5Ce^{3+}(aq)+MnO_4^-(aq)+8H^+(aq)$$

$$MnO_4^-(aq)+8H^+(aq)+5e^-\rightarrow Mn^{2+}(aq)+4H_2O(l);\qquad E^0=+1.51V$$

$$Ce^{4+}(aq)+e^-\rightarrow Ce^{3+}(aq)E^\circ=+1.61V$$

1) – 9.65
2) – 24.3
3) – 48.25
4) – 35.2

20. The standard free energy change for the following reaction is -210 kJ. What is the standard cell potential? $2H_2O_2(aq)\rightarrow 2H_2O(l)+O_2(g)$

1) +0.752
2) +1.09
3) +0.420
4) +0.640

21. The E° at 25°C for the following reaction at the indicated concentrations is 1.50 V. Calculate the AG in kJ at 25°C:

$Cr(s)+3Ag^{+}(aq,0.1M)\rightarrow 3Ag(s)+Cr^{3+}(aq,0.1M)$

1) – 140.94
2) – 295
3) – 212
4) – 422.83

22. If the ΔG of a cell reaction $AgCl+e^{-}\rightarrow Ag+Cl^{-}$ is $-21.20\,kJ;$ the standard emf of the cell is:

1) 0.229 V
2) 0.220 V
3) – 0.220 V
4) – 0.110 V

23. The pressure of H_2 required to make the potential of H_2-electrode zero in pure water at 298 K is:

1) 10^{-10} atm
2) 10^{-4} atm
3) 10^{-14} atm
4) 10^{-12} atm

24. For the reduction of silver ion with copper metal, the standard cell potential was found to be +0.46 V at 25°C. The value of standard Gibbs energy $\left(\Delta G^{0}\right)$ will be: (1F = 96500 C mol^{-1})

1) – 98.0 kJ
2) – 88.7 kJ
3) – 88. 7 J
4) – 44 .5 kJ

25. The thermodynamic efficiency of a cell is given by:

1) $\Delta H / \Delta G$
2) $nEF / \Delta G$
3) $-nEF / \Delta H$
4) nFE^{0}

26. The equilibrium constant of the cell reaction:

$Cu(s)+2Ag^{+}(aq)\rightleftharpoons Cu^{+2}(aq)+2Ag(s)$ is 4×10^{15} then find E^{0}_{cell} at 298 K is approximately

1) 1.21 V
2) 1.15 V
3) 0.46 V

4) 0.85 V

27. For a reaction, $A(s)+2B^{+} \rightleftharpoons A^{2+}+2B(s)$. if the standard electrode potential is 0.354 Volt, then find the equilibrium constant at 298 K

1) 10^{12}
2) 10^{14}
3) 10^{16}
4) 10^{10}

28. If $E^{0}_{Fe^{+2}/Fe}=-0.441\ V$ and $E^{0}_{Fe^{+3}/Fe^{+2}}=0.771$ V the standard EMF of the reaction $Fe+2Fe^{+3} \rightarrow 3Fe^{+2}$ will be:

1) 0.330 V
2) 1.653 V
3) 1.212 V
4) 0.111 V

29. The hydrogen electrode is dipped in a solution of pH = 3 at $25^{0}C$. The electrode potential of the half-cell would be:

1) 0.177 V
2) – 0.177 V
3) 0.087 V
4) 0.059 V

30. $E^{0}\left(Ni^{2+}/Ni\right)=-0.25V$
$E^{0}\left(Au^{+3}/Au\right)=1.50V$

The emf of the voltaic cell. $Ni|\ Ni^{2+}(1.0M)|\ |\ Au^{3+}(1.0M)|\ Au$ is:

1) 1.25 V
2) – 1.75 V
3) 1.75 V
4) 4.0 V

Answer Key

(1)	1	(16)	3
(2)	3	(17)	2
(3)	3	(18)	1
(4)	1	(19)	3
(5)	2	(20)	2
(6)	1	(21)	4
(7)	3	(22)	2
(8)	3	(23)	3
(9)	1	(24)	2
(10)	4	(25)	3
(11)	4	(26)	3
(12)	3	(27)	1
(13)	1	(28)	3
(14)	4	(29)	2
(15)	2	(30)	3